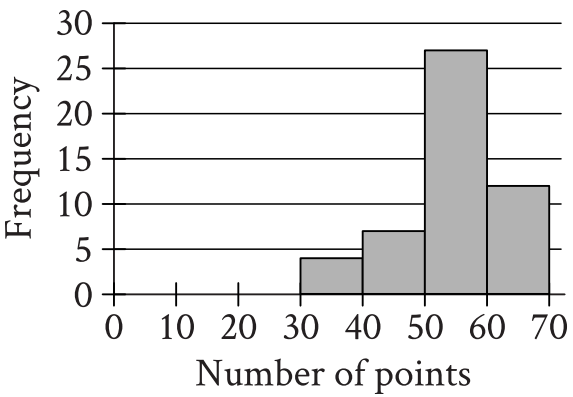


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question



The histogram summarizes data set A, which represents the number of points per player earned by **50** players of a game. A new player earns **18** points playing the game, and this number of points is added to data set A to create data set B with **51** values. Which of the following must be true?

- I. The median number of points per player for data set B is less than the median number of points per player for data set A.
- II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values divided by the number of values. It's given that a new player earns **18** points playing the game, and this number of points is added to data set A to create data set B with **51** values. The histogram shows that all the values in data set A are greater than **18**. Since the additional number of points, **18**, is less than any of the values in the original data set, the mean number of points per player for data set B is less than the mean number of points per player for data set A. The median of a data set with **50** values is between the **25th** and **26th** values when the values are listed in ascending order. The median of a data set with **51** values is the **26th** value when the values are listed in ascending order. Therefore, the median of data set B is the **26th** value in data set B, which is the **25th** value in data set A. The histogram shows that the **25th** and **26th** values in data set A are both between **50** and **60** points. Therefore, the **25th** and **26th** values in data set A could be equal, in which case the median of data set A would be equal to both the **25th** and **26th** values in data set A. Then, the median number of points per player for data set A would be equal to the median number of points per player for data set B. Therefore, only statement II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.